Lecture 8: Fermi Problems and Guesstimation

Based on the lecture slides

Outline

1. What a Fermi problem is
2. Enrico Fermi and order-of-magnitude reasoning
3. A method for estimates
4. Worked examples: Earth, piano tuners, ambulances, and galaxy collisions
5. Using estimates as physics arguments

1 Fermi Problems

A Fermi problem is an estimation problem where exact information is unavailable or unnecessary. The goal is to obtain a quantitative answer within an order of magnitude. This is an important skill in physics because it tells us whether a proposed answer is plausible before we do a detailed calculation.

> The point is not to guess. The point is to break a large question into smaller questions, write clear assumptions, keep units visible, and check whether the final scale makes sense.

2 Enrico Fermi

Enrico Fermi was famous for quick order-of-magnitude estimates. One well-known story concerns the Trinity test in 1945. Fermi estimated the explosive yield by dropping small pieces of paper before and during the passage of the blast wave, measuring their displacement, and relating that displacement to the pressure of the shock.

The historical details are less important for us than the method: simple observations, simple physics, and a controlled approximation can already tell us a lot.

3 A Practical Method

When solving a Fermi problem:

1. Break the problem into smaller pieces.

2. Use quantities you roughly know. If you do not know them directly, estimate them from something nearby.
3. Round aggressively, but not always in the same direction.
4. Use scientific notation.
5. Keep units attached to every intermediate result.
6. Check the final result against common sense or known scales.

Typical useful approximations are

$$\pi \simeq 3, \qquad c \simeq 3 \times 10^8\,\mathrm{m/s}, \qquad 1\,\mathrm{mile} \simeq 1.6\,\mathrm{km}. \tag{1}$$

4 Example 1: Diameter of the Earth

Estimate the diameter of the Earth from the distance between New York and Los Angeles.

Assumptions

- New York and Los Angeles are separated by about three time zones.
- The distance between them is about 5000 km.
- There are 24 time zones around the Earth.

The distance per time zone is therefore roughly

$$\frac{5000\,\mathrm{km}}{3} \simeq 1700\,\mathrm{km}. \tag{2}$$

The circumference of the Earth is then

$$C_\oplus \simeq 24 \times 1700\,\mathrm{km} \simeq 4 \times 10^4\,\mathrm{km}. \tag{3}$$

Since $C_\oplus = 2\pi R_\oplus$,

$$R_\oplus \simeq \frac{4 \times 10^4\,\mathrm{km}}{6} \simeq 6.5 \times 10^3\,\mathrm{km}, \tag{4}$$

and

$$D_\oplus \simeq 1.3 \times 10^4\,\mathrm{km}. \tag{5}$$

The actual value is about 12742 km.

5 Example 2: Piano Tuners in Chicago

Estimate the number of piano tuners in Chicago.

Assumptions

- Chicago has about 2×10^6 people.
- There are about 4 people per household.
- About 1 household in 20 has a piano.
- Each piano is tuned once per year.
- A tuning takes about 2 hours.
- A tuner works about 8 hours/day, 5 days/week, and 50 weeks/year.

The number of pianos is

$$N_{\text{piano}} \simeq 2 \times 10^6 \left(\frac{1\,\text{household}}{4\,\text{people}}\right)\left(\frac{1\,\text{piano}}{20\,\text{households}}\right) \simeq 2.5 \times 10^4. \tag{6}$$

So Chicago needs roughly 2.5×10^4 tunings per year. A tuner can do

$$\left(2000\,\frac{\text{hr}}{\text{yr}}\right)\left(\frac{1\,\text{tuning}}{2\,\text{hr}}\right) \simeq 10^3\,\frac{\text{tunings}}{\text{yr}}. \tag{7}$$

Therefore,

$$N_{\text{tuners}} \simeq \frac{2.5 \times 10^4}{10^3} \simeq 25. \tag{8}$$

6 Example 3: Ambulance Stations in the Netherlands

Estimate how many ambulance stations are needed if an ambulance must be able to reach an emergency within about 15 minutes.

Assumptions

- Approximate the Netherlands by a rectangle of width 200 km and height 300 km.
- The area is therefore $A \simeq 6 \times 10^4\,\text{km}^2$.
- Because of response time, use about 12 minutes of driving.
- The average ambulance speed is about 50 km/hr.
- Each station covers roughly a circle of radius r.

The reachable radius is

$$r \simeq 50\,\frac{\text{km}}{\text{hr}}\left(\frac{12\,\text{min}}{60\,\text{min/hr}}\right) \simeq 10\,\text{km}. \tag{9}$$

The area covered per station is approximately

$$B \simeq \pi r^2 \simeq 3 \times 10^2\,\text{km}^2. \tag{10}$$

Thus the number of stations is

$$N \simeq \frac{A}{B} \simeq \frac{6 \times 10^4}{3 \times 10^2} \simeq 2 \times 10^2. \tag{11}$$

This is close to the actual scale, which is why this is a good Fermi problem: simple assumptions already give the right order of magnitude.

7 Example 4: Galaxy Collisions

If two galaxies collide, how many individual stars are expected to collide?

Assumptions

- A Milky-Way-like galaxy contains about 10^{11} stars.
- The galaxy has radius $R_{\rm gal} \simeq 5 \times 10^4$ light-years.
- One light-year is about 10^{16} m, so $R_{\rm gal} \simeq 5 \times 10^{20}$ m.
- The typical stellar radius is of order the solar radius, $R_\star \simeq 10^9$ m.

The volume of the galaxy is roughly

$$V_{\rm gal} \simeq 3R_{\rm gal}^3 \simeq 3(5 \times 10^{20}\,\mathrm{m})^3 \sim 5 \times 10^{62}\,\mathrm{m}^3. \tag{12}$$

The number density of stars is then

$$n_\star \simeq \frac{10^{11}}{5 \times 10^{62}} \sim 10^{-52}\,\mathrm{m}^{-3}. \tag{13}$$

During a head-on passage through another galaxy, a star sweeps out a volume

$$V_{\rm swept} \sim R_{\rm gal} \times 3R_\star^2 \sim (5 \times 10^{20})(3 \times 10^{18}) \sim 10^{39}\,\mathrm{m}^3. \tag{14}$$

The collision probability for one star is

$$P \sim n_\star V_{\rm swept} \sim 10^{-52}10^{39} \sim 10^{-13}. \tag{15}$$

Multiplying by 10^{11} stars gives far less than one direct stellar collision per galaxy collision, at most of order 10^{-2} to 10^{-1} with these crude assumptions. Galaxies can pass through each other while individual stars almost never hit.

8 Physical Meaning: The Bullet Cluster

The fact that stars rarely collide in galaxy-cluster mergers is physically useful. In systems such as the Bullet Cluster, galaxies pass through one another with little direct stellar collision, while gas does collide and emits X-rays. Gravitational lensing shows that most of the mass follows the collisionless component rather than the gas. This is one of the classic pieces of evidence for dark matter.

9 Handy Scales

The following approximate scales are useful for quick estimates:

- Earth diameter: 10^7 m.
- Human height: 10^0–10^1 m.
- Sugar cube: 10^{-2} m.
- Atom: 10^{-10} m.
- Proton: 10^{-15} m.
- Water density: 10^3 kg/m^3.
- Human mass: 10^2 kg.
- Earth mass: 10^{25} kg.
- Sun mass: 10^{30} kg.

Exercises

1. Estimate how many breaths you take in a year.
2. Estimate the mass of air in this lecture room.
3. Estimate the number of photons emitted per second by a 100 W light bulb, assuming visible photons have energy of a few eV.
4. Estimate how many relativistic muons created in the upper atmosphere reach sea level, and identify where special relativity enters the estimate.